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Don't Put All Your Eggs in One Cloud: Lessons Learned from the October 2025 AWS Outage

Posted by INE



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ts across the world started receiving



larger: AWS US-EAST-1, the crown jewel of Amazon's cloud infrastructure and home to countless production workloads, was experiencing a catastrophic failure.

For the next 15 hours, businesses that had built their entire infrastructure in this region watched helplessly as their services remained offline. E-commerce sites couldn't process orders. SaaS platforms couldn't serve customers. Mobile apps couldn't reach their backends. The "highly available" cloud that promised resilience had become completely unavailable.

Many engineers had done everything "right." They deployed applications across multiple availability zones, implemented health checks, and followed the AWS Well-Architected Framework. Unfortunately, none of it mattered when the entire region failed.

Root Cause: AWS DNS Failure and Service Cascade

The root cause was deceptively simple: DNS resolution for DynamoDB service endpoints began to fail within AWS's internal network. In distributed systems, though, simple failures rarely remain that way.

When internal DNS queries stopped working, a cascade began. EC2, AWS's elastic compute service, depends on DynamoDB services for internal operations. Without this, new EC2 instance launches started failing region-wide. This triggered Network Load Balancer health check failures, creating widespread connectivity issues that affected Lambda, CloudWatch, and dozens of other AWS services.

The critical detail here is that this was a DNS problem internal to AWS, and not related to Route 53 (Amazon's DNS hosting service), or any other customer-side DNS services. This means that once AWS's own services couldn't find each other, it was out of their customers' hands. Workloads in US-EAST-1 were effectively stranded, and even deployments across all six availability zones went down together.

The businesses that stayed online had made one key architectural decision: they weren't

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True Cloud Resilience



guarantee 99.99% (four nines) for most services, allowing 52.6 minutes of annual downtime. Some services only guarantee 99.95%.

This outage lasted 15 hours, obliterating that annual budget by a factor of 17. A mid-size e-commerce site processing \$100,000 daily would lose \$62,500 in revenue during this outage, not counting recovery costs and damaged customer trust.

Amazon's US-EAST-1 region hosts an estimated 30-40% of all AWS workloads globally. In fact, the surrounding Northern Virginia corridor hosts the densest concentration of data center infrastructure globally, a hub through which an estimated 70% of the world's Internet traffic flows. Unfortunately, this region's offer of cheap and fast connectivity also comes with a significant increase in risk. This stems from the fact that Availability Zones can protect against data center infrastructure failures - like a power outage or a network issue within a single facility - but they don't protect against region-wide failures affecting the entire control plane, or internal services like DNS resolution that all AZs depend on.

Think of it this way: AZs are like rooms in a house. If one room floods, you move to another. But if the entire house floods, every room is underwater. Regions are separate houses. You need multiple houses for true redundancy.

Three Proven Strategies to Surviving a Cloud Outage

Path 1: Multi-Region Architecture (The Simplest Solution)

Deploying across multiple AWS regions keeps you within one cloud provider while eliminating single points of failure. The minimum viable setup requires:



Primary region (e.g., US-EAST-1)

Secondary region (e.g., US-WEST-2 or a geographically distant region)

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Active-Active: Both regions serve production traffic simultaneously. Traffic splits based on geography or load. When one fails, the other absorbs everything. This offers instant failover but doubles infrastructure costs and requires complex data synchronization.

Active-Passive: Primary region handles all traffic while secondary runs minimal infrastructure, ready to scale. Database replication keeps data current. During failure, traffic redirects to secondary. This balances cost (30-50% increase) with reasonable recovery time.

Pilot Light: Secondary region contains only critical components with minimal capacity. During failure, automation rapidly provisions full infrastructure. Lowest cost option but requires 30-60 minutes for recovery and bulletproof automation scripts.

During the October outage, organizations with multi-region architectures experienced a minor incident instead of a crisis. Their monitoring detected US-EAST-1 issues, traffic failed over automatically, and customers never noticed.

Path 2: Multi-Cloud Strategy (Maximum Protection)

Running workloads across AWS, Google Cloud, and/or Azure provides true provider independence. When AWS has a bad day, your services run elsewhere. This requires more expertise but eliminates provider-level single points of failure.

The reality check: To properly implement a Multi-Cloud solution, your team needs expertise across multiple platforms. Network complexity increases significantly, as connecting AWS VPCs to Google Cloud or Azure networks requires careful planning, and data egress charges between clouds can quickly add up.

If a full Multi-Cloud solution seems overwhelming, consider a selective approach. For example, run authentication, payment processing, and core APIs across clouds. Then keep internal tools and non-critical services on one provider. This protects revenue-generating services without a full doubling of operational overhead.

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is in your own Data Center. Hybrid-
th cloud services, giving you complete



Automatic Failover Saves You from Disaster

Regardless of which path you go down - multiple regions in a single cloud provider, multiple cloud providers, or a hybrid on-prem solution - automating failover is a must. By the time you manually detect, assess, decide, and execute, hours may have already passed during a downtime event. Automatic failover turns regional disasters into minor incidents.

The difference is stark in real numbers. During the October outage, organizations with manual failover processes typically took 2-4 hours just to confirm the scope of the problem, convene decision makers, and approve the failover. Another 1-2 hours to execute the failover and verify services were running in the backup region. That's 3-6 hours of downtime that could have been 3-6 minutes with automation.

Effective automatic failover requires three components working together:



Continuous health monitoring that can distinguish between minor hiccups and actual failures

Predetermined thresholds that trigger failover without human intervention

Well-tested automation that executes the switch reliably

Each component must be sophisticated enough to prevent false positives (failing over when you shouldn't) while remaining sensitive enough to catch real failures quickly.

Critical Professional Skills for the Multi-Cloud Era

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age to anyone building cloud-based
no longer optional specializations, they're

Shift Your Learning Strategy:



Azure, and GCP. Start with your primary cloud, then learn how equivalent services work elsewhere.

Focus on portable technologies. Kubernetes, Terraform, and Ansible work everywhere. Proprietary services like Lambda or DynamoDB lock you in. Learn both, but prioritize portable skills that transfer across providers.

Master networking fundamentals. DNS, BGP, VPNs, and load balancing concepts apply everywhere. Understanding core networking helps you diagnose problems regardless of provider.

INE provides extensive training for professionals aiming to build robust networking and cloud infrastructures. Our learning paths offer hands-on training, preparing engineers to mitigate and minimize downtime during future outages. This includes expertise in implementing automated failover, mastering portable technologies, and understanding core networking fundamentals.

Start training with these learning paths and courses:



Cloud Networking for Network Professionals

AZ-900: Azure Fundamentals

Kubernetes From Scratch

Introducing Terraform for the Network Engineer

Configuration Consistency with Ansible

BGP for the Enterprise

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in their US-EAST-1 region, those were 15 hours of helplessness. For organizations with

against Outages

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Cloud failures will happen again. Every major provider has experienced them. The question isn't whether you'll face another outage, but whether your architecture will be ready.

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